You can use that `visit` function to do more than optimized search. I've used it to cluster nearby points on a map (http://bl.ocks.org/emeeks/066e20c1ce5008f884eb) and also to draw the bounds of the quadtree in figure 11.13.

11.4 More optimization techniques

You can improve the performance of the data visualization of large datasets in many other ways. Here are three that should give you immediate returns: avoid general opacity, avoid general selections, and precalculate positions.

11.4.1 Avoid general opacity

Whenever possible, use fill-opacity and stroke-opacity or RGBA color references rather than the element opacity style. General element opacity, the kind of setting you get when you use `"style: opacity"`, can slow down rendering. When you use specific fill or stroke opacity, it forces you to pay more attention to where and how you're using opacity.

So instead of

```
d3.selectAll(elements).style("fill", "red").style("opacity", .5)
```

do this:

```
d3.selectAll(elements).style("fill", "red").style("fill-opacity", .5)
```

11.4.2 Avoid general selections

Although it's convenient to select all elements and apply conditional behavior across those elements, you should try to use `selection.filter` with your selections to reduce the number of calls to the DOM. If you look at the code in listing 11.16, you'll see this general selection that clears the selected attribute for all the circles and sets the fill of all the circles to pink:

```
d3.selectAll("circle")
.style("fill", "pink")
.each(function(d) {d.selected = false})
```

Instead, clear the attribute and set the fill color of only those circles that are currently set to the selection. This limits the number of costly DOM calls:

```
d3.selectAll("circle")
.filter(function(d) {return d.selected})
.style("fill", "pink")
.each(function(d) {d.selected = false})
```

If you adjust the code in that example, the performance is further improved. Remember that manipulating DOM elements, even if it's changing a setting like fill, can cause the greatest performance hit.